

Top-K Videos

Functional requirements:

user should be able to view top-K videos,
top-K should be sorted accordingly to user's activity,
top-K videos should be updated after every 5seconds

Non-Functional requirements:

data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime , i.e
Availability>>Consistency)

Core Entities:

user,
videos,
history

API :

POST:upload/video
{
 userID,
 {filemetadata}
}
GET:top/{K}

HLD :

